



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Emergence of Big Data

Categorizing software systems accurately and crafting a complete list of all possible categories is almost impossible.

However, we can still start with the following categories listed below.

Software Engineering Specializations

1. System Software
3. Programming Tools
3. Desktop Applications
4. Data Processing Application
5. Web & Mobile Applications
6. Software Platforms
7. Cloud Computing
8. Machine Learning and AI

We saw the evolution of the COBOL programming language specifically designed for business data processing.

The COBOL, also known as Common Business-Oriented Language, was the first of its kind.

COBOL allowed us to store data in files, create index files, and process data efficiently.

However, we saw data processing shift from COBOL to relational databases such as Oracle and Microsoft SQL Server.

Beginning of Business Data Processing

COBOL

- Common Business-Oriented Language
- Used in business and finance for companies and governments
- Allowed to store data in files, create index files, and process data efficiently
- Widely used on mainframe computers
- Used for large-scale batch and transaction processing jobs

Out of all the software specializations we have listed earlier, I wanted to attract your attention towards Data processing applications.

You can think of COBOL as the first serious attempt towards enabling data processing. And COBOL was designed in 1959.

The Oracle database achieved the subsequent major success in enabling data processing. And Oracle was founded in 1977.

So data processing has always been at the centre of the Software industry. Everything else will come and go, but data will only grow.

Data processing is an evergreen field for software engineers.

Technologies may evolve, but the requirement for processing more and more data at a faster speed will keep becoming more critical.

We have used RDBMS technology for many decades.

Some popular RDBMS systems are Oracle, SQL Server, PostgreSQL, MySQL, Teradata, and Exadata.

These RDBMS systems offered us three main features to help us develop Data Processing applications.

1. SQL - An easy Data Query Language
2. Scripting Languages such as PL/SQL and Transact SQL
3. Interface for other programming languages such as JDBC and ODBC

So we used SQL for querying data and PL/SQL for doing things that we couldn't do using SQL.

They also offered interfaces such as ODBC/JDBC so we could interact with data using the programming languages.

We could create data processing applications using these technologies.

We had the following requirements in the Big Data problem.

Big Data Platform Requirements



Store High Volumes of data
arriving at a higher velocity



Accommodate structured, semi-
structured, and unstructured
data variety

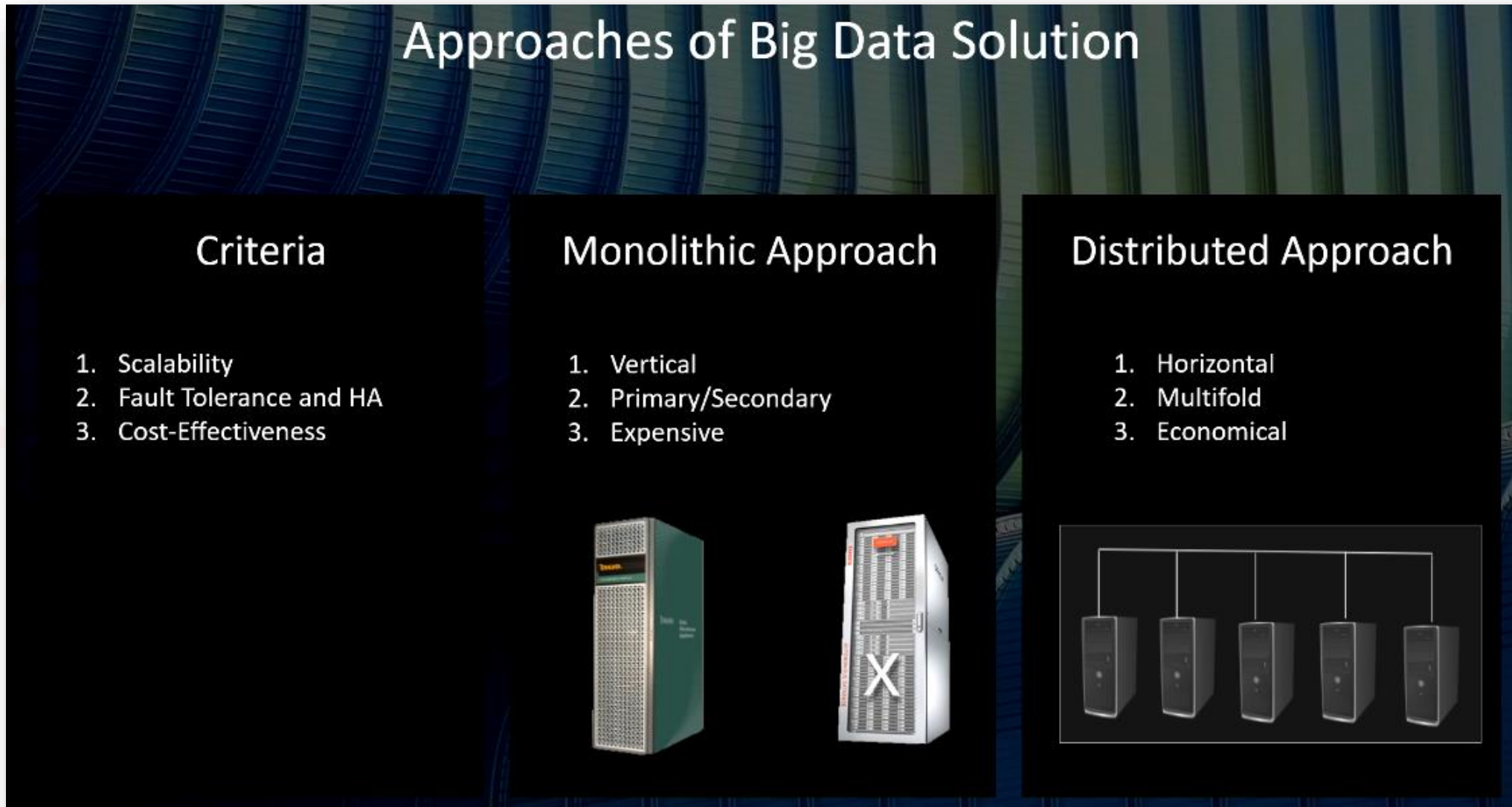


Process high volumes of a
variety of data at a higher
velocity

There was two approaches to the big data problem:

1. The monolithic approach designs one large and robust system that handles all the requirements. Teradata and Exadata are examples. These two systems mainly support only structured data. So we cannot call them big data systems, but they are designed using a monolithic approach.
2. The distributed approach takes many smaller systems and bring them together to solve a bigger problem.

Here are the two approaches to Big Data solution.



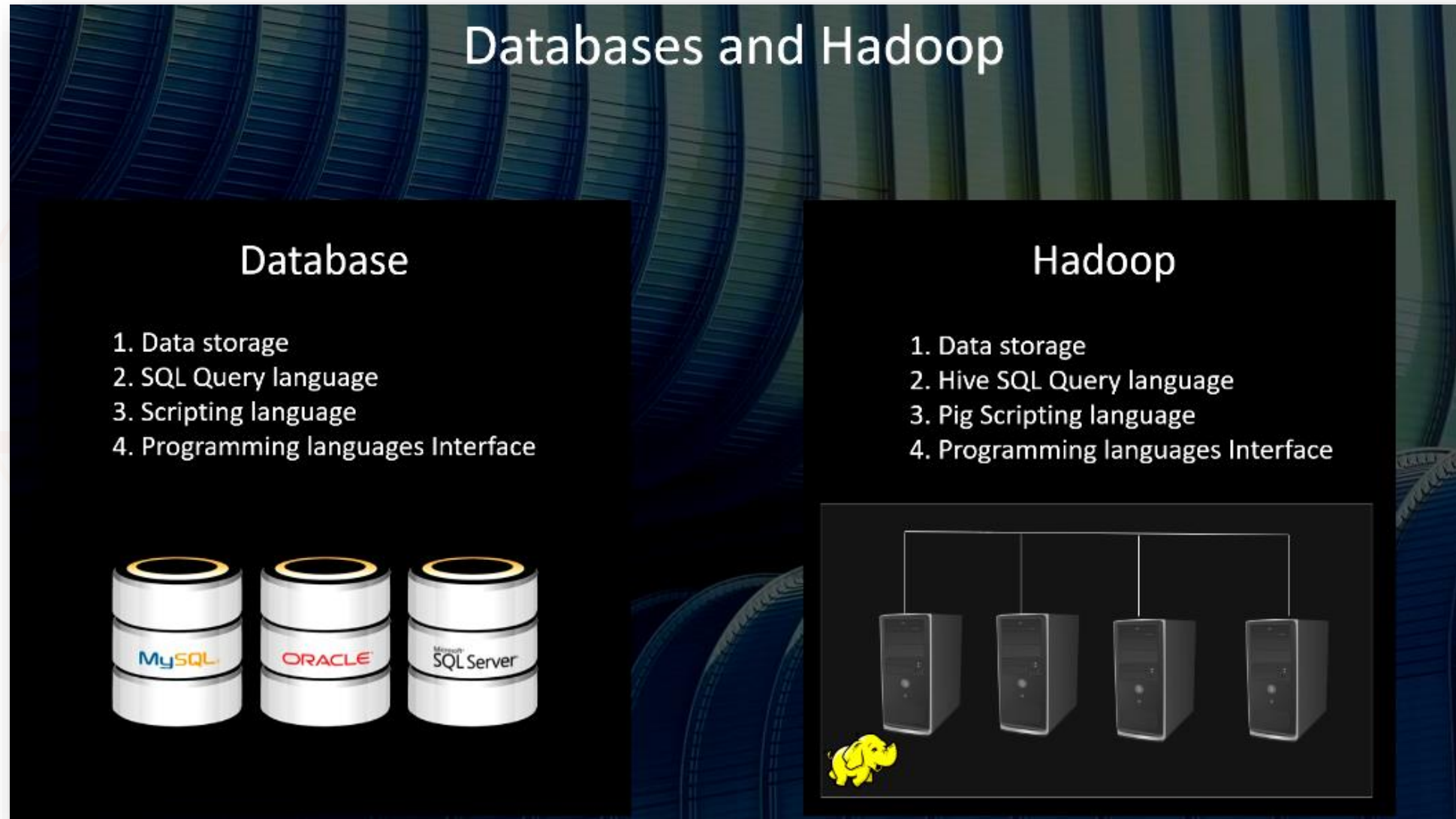
Hadoop came up as a new data processing platform to solve Big Data problems.

The Hadoop platform was designed and developed in layers.

The core platform layer offered three capabilities:

1. Distributed cluster formation or Cluster Operating System
2. Data storage and retrieval on the distributed cluster or Distributed Storage
3. Distributed data processing using Java programming language or Map-Reduce Framework

Here is a comparison between Database and Hadoop.





Thank You
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